

Lab 2: measuring a single pendulum

From high-speed video to a defensible model comparison

Physical model, finite-angle behavior, camera measurement, tracking, and Pre-Lab 2.



What you will do in Lab 2

Release pendulum from rest at three angle ranges.

Record motion with high-speed camera.

Track bob and convert $x(t)$, $y(t)$ into $\theta(t)$.

Compare period and frequency with model.



Physical system and coordinate

M: bob and attached hardware mass

m: uniform bar mass

l: pivot-to-bob-center distance

$\theta(t)$: angle from downward vertical

$\omega(t) = d\theta/dt$; every release begins with $\omega(0) = 0$.

Ideal point-mass pendulum

$$M\ell^2 \, d^2\theta/dt^2 = -Mg\ell \sin \theta$$

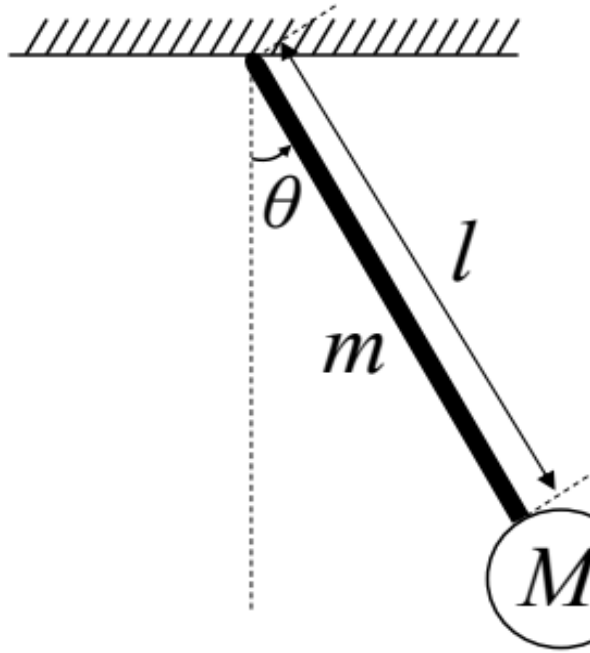
$$d^2\theta/dt^2 + (g/\ell) \sin \theta = 0$$

Gravity drives the bob toward $\theta = 0$.

The sine term makes exact motion nonlinear.

M cancels only because this ideal model treats the bar as massless.

The bar mass changes the model



Moment of inertia about the pivot

$$I_O = M\ell^2 + \frac{1}{3}m\ell^2$$

Gravity torque

$$\tau_g = -(Mg\ell + \frac{1}{2}mg\ell) \sin \theta$$

The bob acts at l . The uniform bar acts through its center of mass at $l/2$.

Compound-pendulum equation

$$(M\ell^2 + \frac{1}{3}m\ell^2) d^2\theta/dt^2 + (Mg\ell + \frac{1}{2}mg\ell) \sin \theta = 0$$

Small-angle approximation

$$\omega_n^2 = g(M + m/2) / [\ell(M + m/3)]$$

$$f_n = \omega_n/(2\pi) \quad T_0 = 2\pi/\omega_n$$

Measure M , m , and ℓ for the apparatus.

The point-mass formula applies only when bar mass is negligible.

Mass distribution changes inertia and gravity torque by different amounts.

Why large initial angle changes period

At large angle, $\sin(\theta)$ differs from θ .

Exact restoring torque is weaker than the linear model predicts at the same angle.

Finite-angle motion takes longer to complete a cycle.

Use the nonlinear equation from Pre-Lab 2 for all three releases.



Energy during a swing

$$E = \frac{1}{2}(M\ell^2 + \frac{1}{3}m\ell^2)(d\theta/dt)^2 + (Mg\ell + \frac{1}{2}mg\ell)(1 - \cos \theta)$$

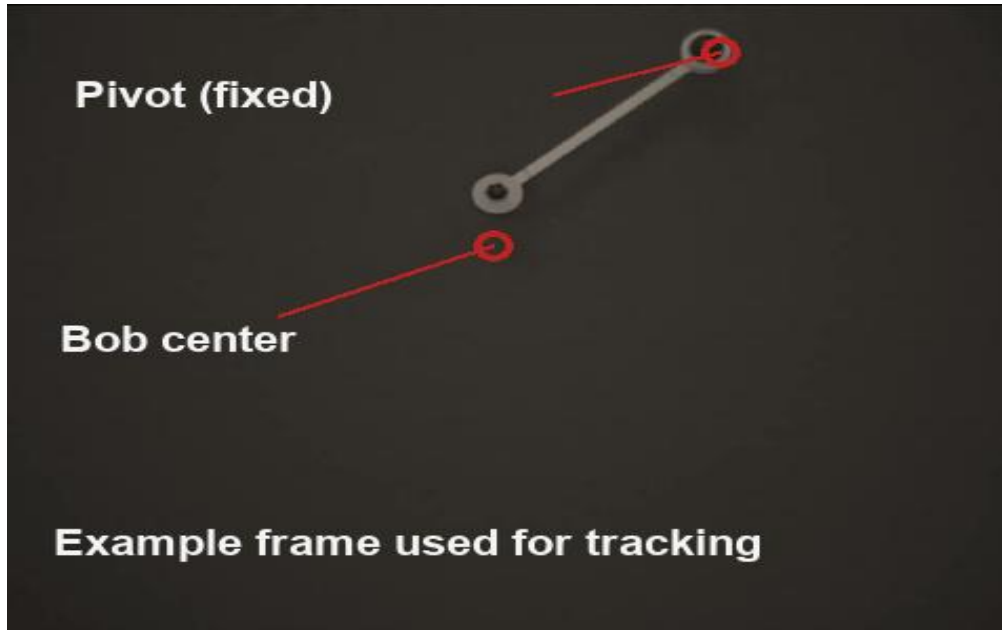
At release theta is largest and omega = 0.

Near the bottom theta = 0 and speed is largest.

Air drag and pivot friction cause measured amplitude and energy to decrease.

What the camera actually measures

What is recorded: a video frame



The frame must show the fixed pivot and the bob center.

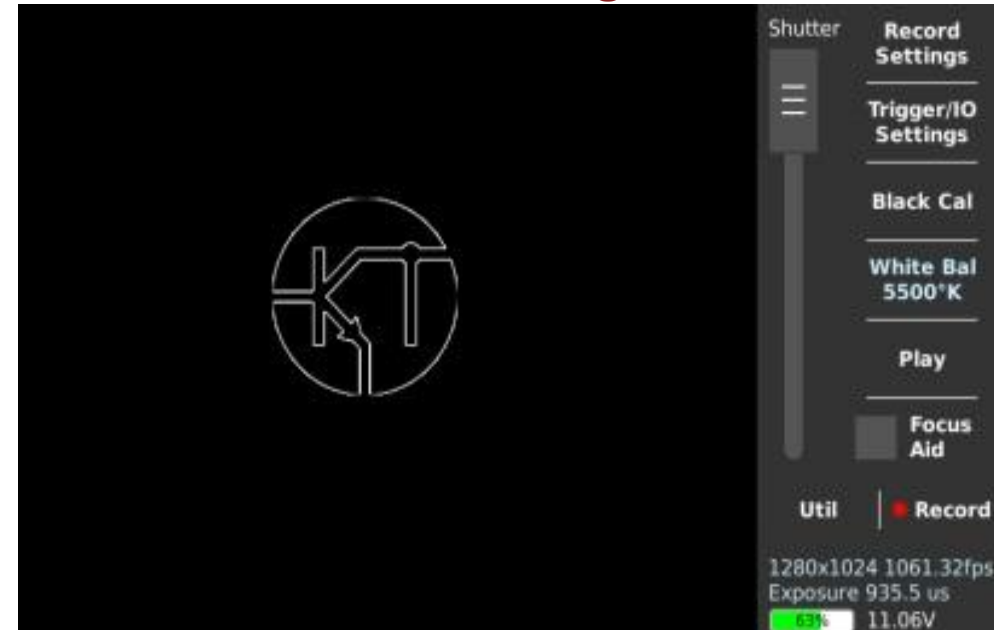
The camera stores frames, not angle directly.

For every frame, the analyzer finds the pivot and bob-center positions.

Those two positions give $\theta(t)$; the chosen frame rate gives the time step between frames.

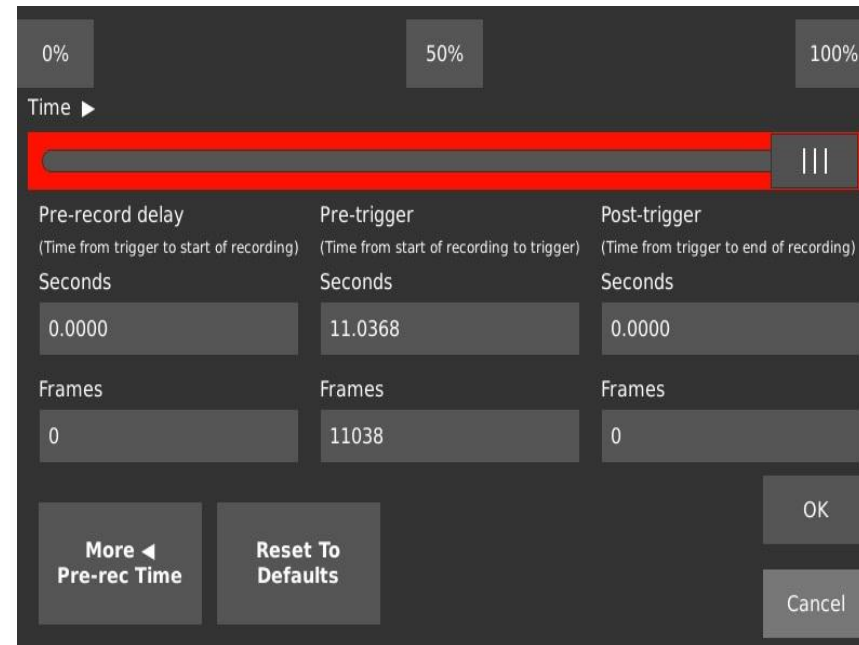
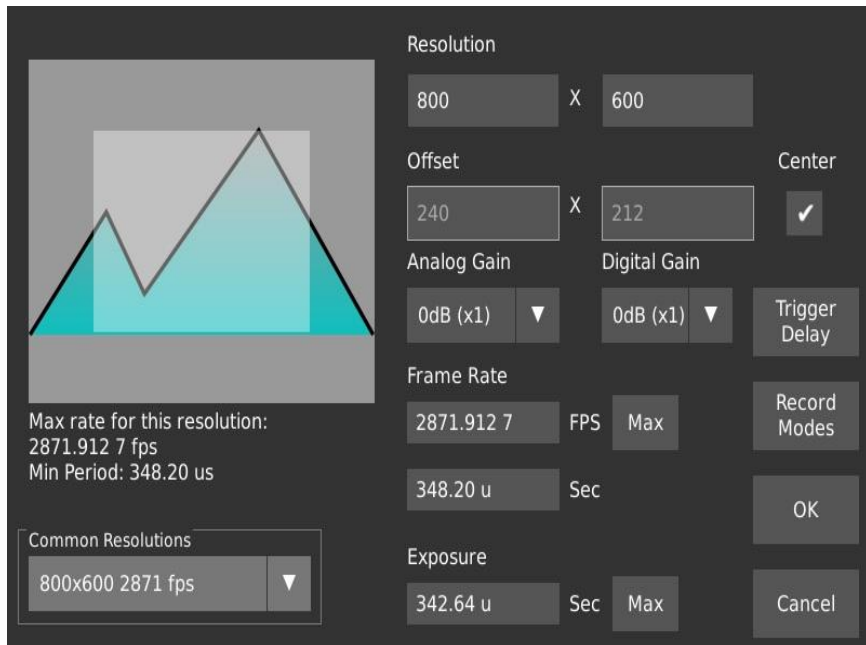
Keep full motion in frame and use sharp contrast so the center can be tracked.

What controls the recording



Camera interface: frame rate, exposure, focus, and Record.

Camera settings and useful video



Record the actual frame rate and resolution shown by the camera; do not copy a number from this example screen.

Frame rate f_s sets the time spacing: $\Delta t = 1/f_s$. Exposure/shutter controls blur. Playback rate only changes viewing speed.

Use trigger delay that includes release and several complete cycles.

Test framing before experiment. Smooth video does not prove accurate tracking.

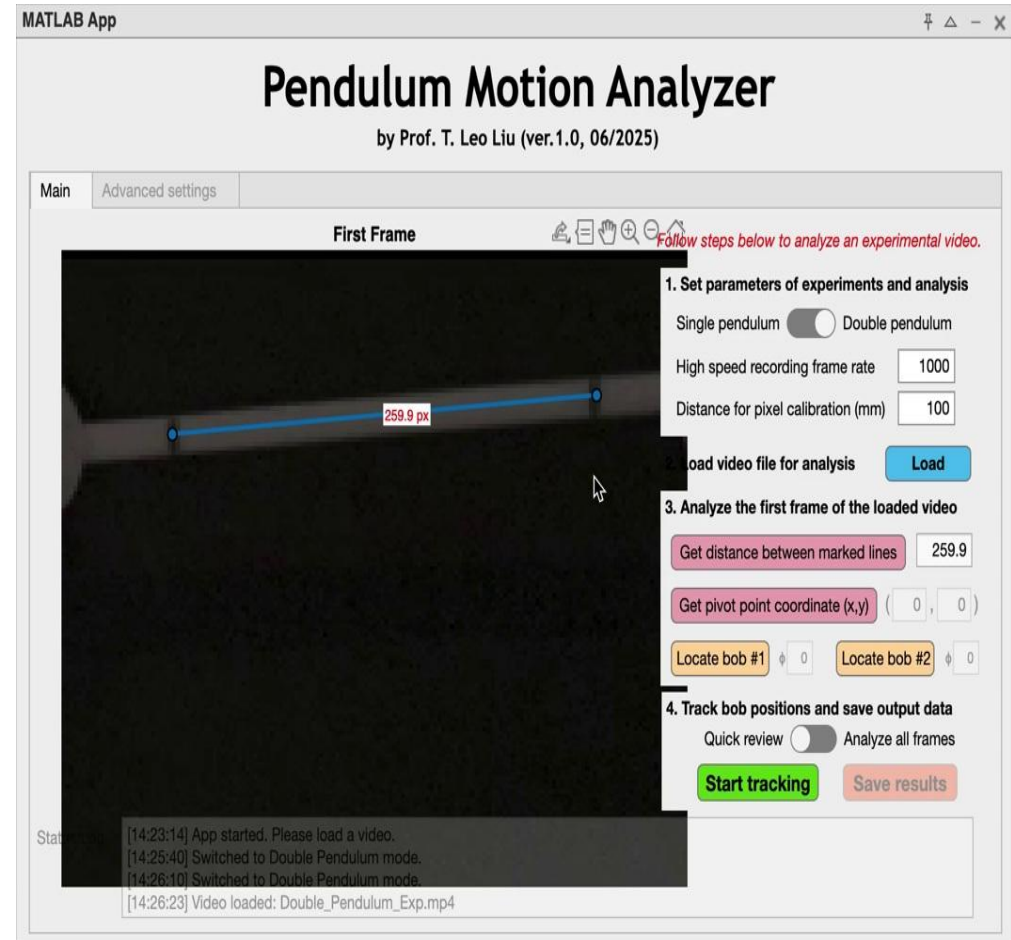
From pixels to coordinates

Calibration marks have known separation, normally 100 mm.

Calibration gives pixel-to-millimeter scale.

Do not move camera or change zoom after calibration.

App saves time and bob coordinates.



From coordinates to a signed angle

$$\theta(t) = \text{atan2}(x_b - x_p, y_b - y_p)$$

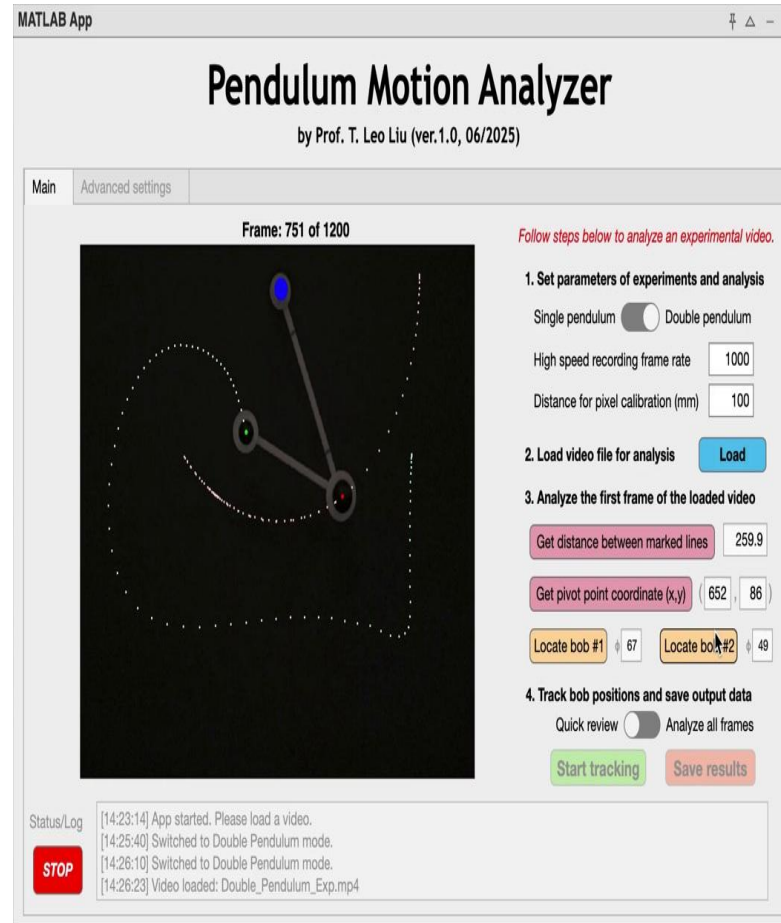
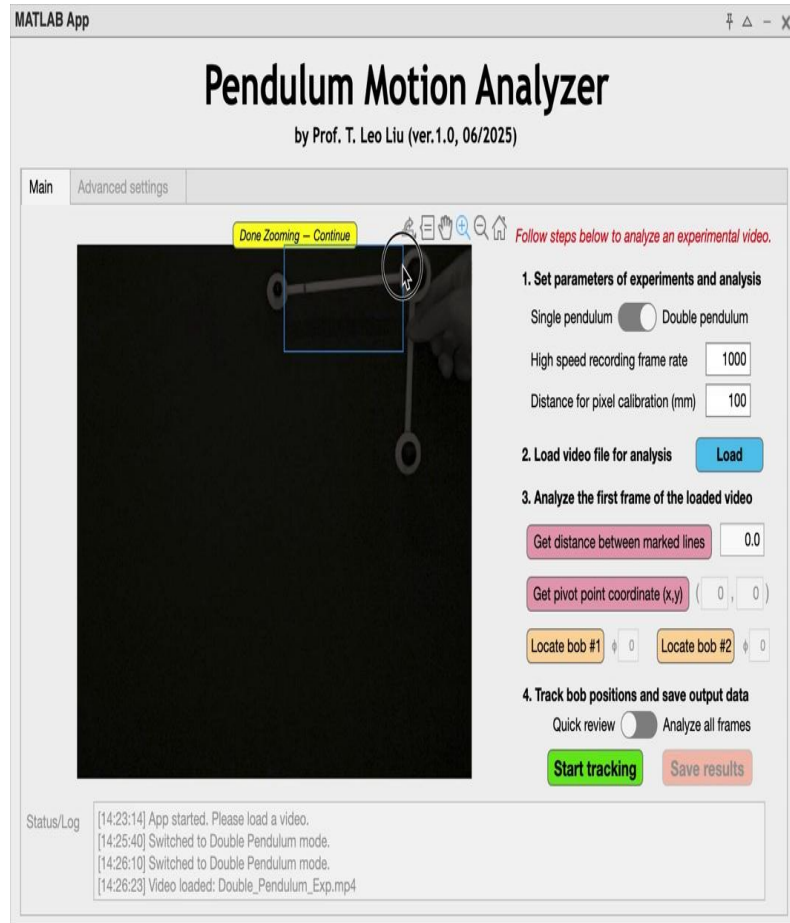
For image coordinates, x points right and y points down.

The formula gives $\theta = 0$ when the bob is directly below the pivot.

Verify zero and sign using one known frame before batch analysis.

Use radians for modeling and derivatives.

Tracking workflow in MATLAB



Select Single pendulum.

Enter actual camera frame rate.

Calibrate, select pivot and bob, then Quick Review.

Run Analyze All Frames only after marker follows bob.

Why frame rate belongs in report

$$f_{\text{Nyquist}} = f_s/2 = 500 \text{ Hz} \quad (f_s = 1000 \text{ frames/s})$$

For this slow pendulum, the assigned high-speed rate provides many frames per cycle.

The actual value of f_s becomes the time axis of the data: $\Delta t = 1/f_s$.

Wrong f_s rescales time, frequency, and angular velocity. Record the value shown by the camera.

Angular velocity and noise

$$\omega(t) = d\theta(t)/dt$$

Differentiation magnifies frame-to-frame error.

Inspect $\theta(t)$ before interpreting $\omega(t)$.

State smoothing method and show it does not hide tracking problems.

Maximum speed occurs near the bottom.

Period from time history

Measure several cycles between same-direction peaks in $\theta(t)$.

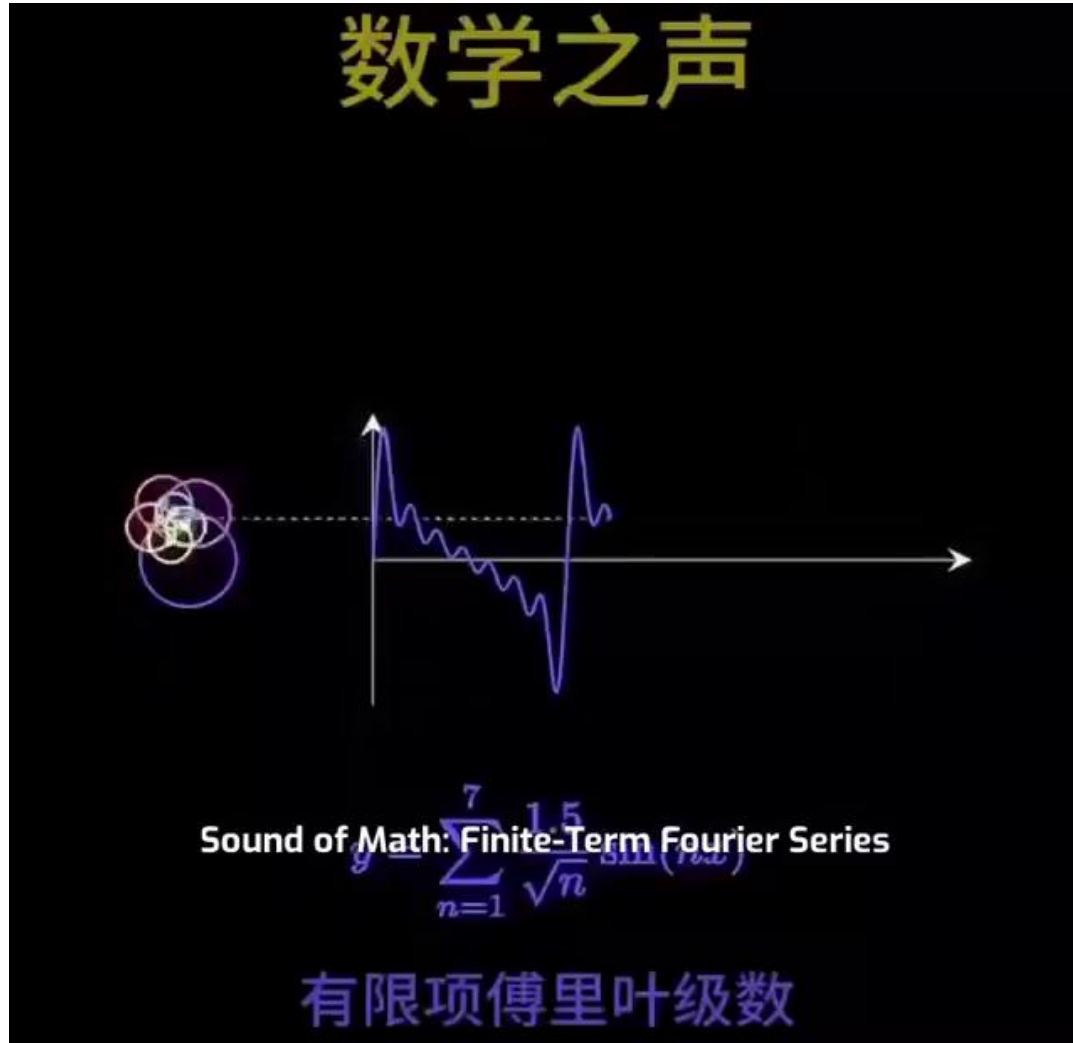
Use $T = (t_{k+n} - t_k)/N$, then $f = 1/T$.

This is the direct physical measurement: “how long does one swing take?”

Use the FFT as an independent frequency-based check.

$$T = (t_{k+n} - t_k)/N \quad f = 1/T$$

What this video shows: Fourier synthesis



Watch the embedded 29-second video

The colored circles are rotating vectors (phasors), each representing one sinusoidal component with its own amplitude, frequency, and phase.

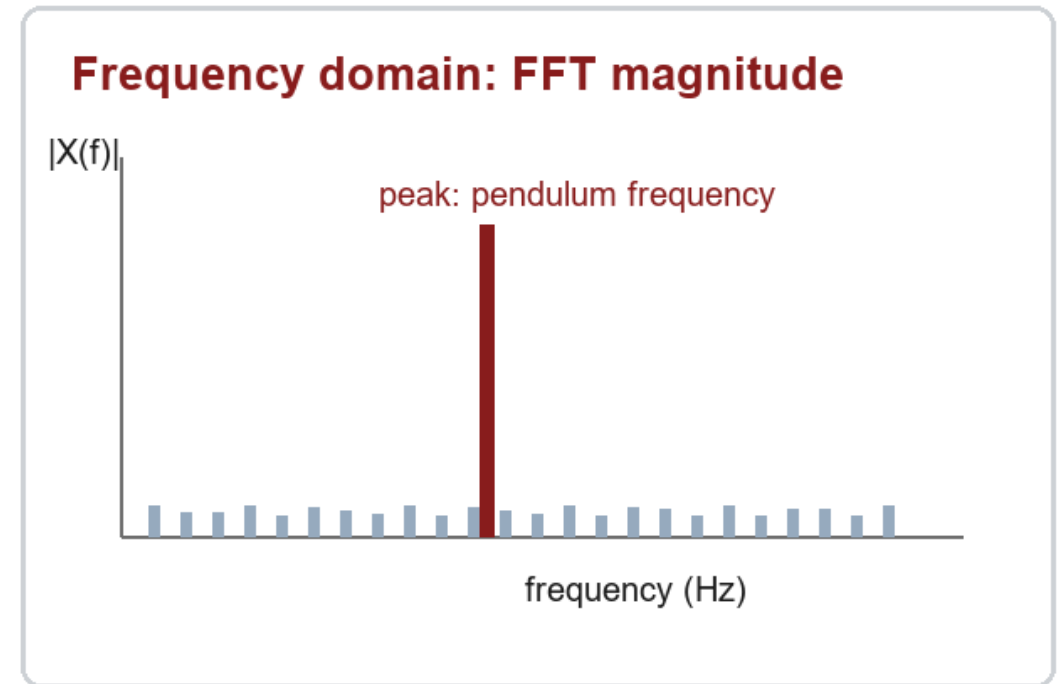
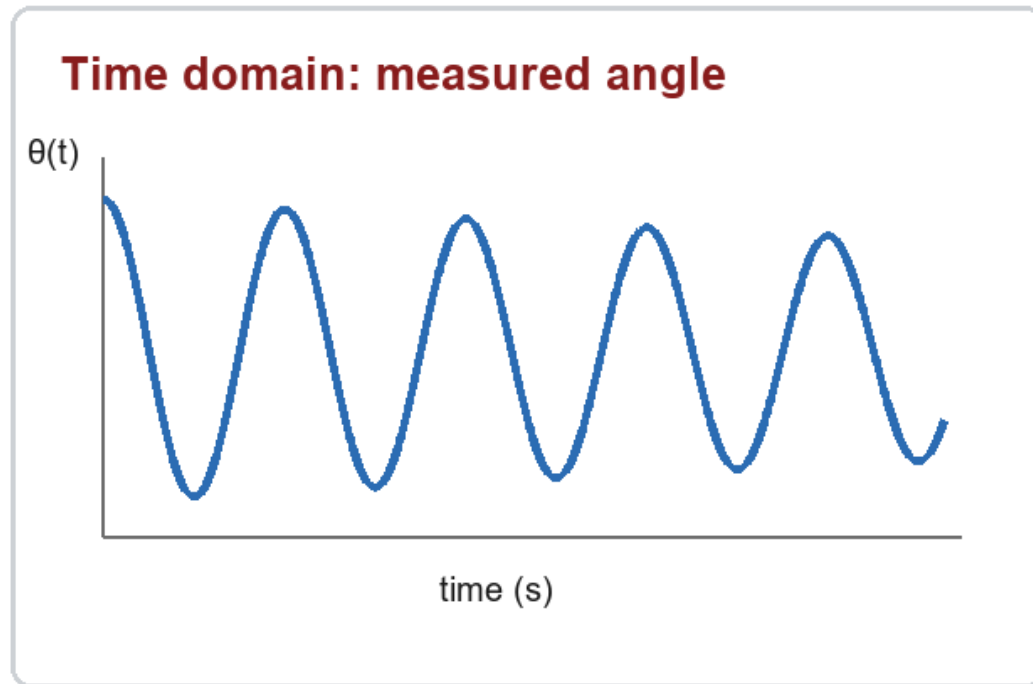
The trace at right is their instantaneous sum. The video therefore builds a waveform from known components: this is Fourier synthesis or a Fourier series.

It is not an FFT calculation and it is not pendulum data. Use it as a visual model for the phrase “a signal can be represented by sinusoidal components.”

In Lab 2 we do the reverse: we start with measured $\theta(t)$, and the FFT estimates the components that best describe that record.

What the FFT means physically

One motion record, viewed in two ways



Same physical motion: the trace shows how angle changes; the peak identifies how often it repeats.

The FFT asks: how strongly does the measured $\theta(t)$ contain a sinusoid at each candidate frequency?

A tall peak near the pendulum frequency means that oscillation is the dominant repeating motion.

The FFT does not create a new physical tone; it reorganizes the same recorded motion by frequency.

Why the pendulum produces an FFT peak

Example: one complete swing every 2 s gives a frequency of 0.5 Hz.

For a small release, the angle is approximately a sinusoid:

$\theta(t) = A \cos(2\pi \times 0.5t + \phi)$. This is an illustrative example, not a measured result.

The DFT measures how much of each candidate sinusoid is present.

The FFT computes those same comparisons efficiently.

The spectrum has its main peak near 0.5 Hz because that sinusoid matches the repeated motion. Check it against the period measured between peaks.

Matching frequency makes the products add up

Multiply the angle samples by a reference cosine and add the products. For an aligned reference, positive $\times$ positive and negative $\times$ negative both add positively.

At a different frequency, the reference drifts in and out of step. Positive and negative products largely cancel over a sufficiently long record.

Use BOTH a sine and a cosine reference at each frequency. Their combined magnitude detects the component regardless of its initial phase.

Peak position tells us the repetition frequency. Peak height depends on component amplitude and spectrum scaling, not the largest angle at one instant.

What FFT measures in the sound experiment

Sound pressure moves the microphone diaphragm. The microphone and acquisition system produce sampled voltage $V(t)$.

One pure 440 Hz tone gives a main spectral peak near 440 Hz.

Two tones at 440 and 700 Hz give two main peaks if the record resolves them.

A musical note may include a fundamental and higher harmonics. Their relative amplitudes help explain why instruments sound different.

The horizontal axis is frequency in Hz. A correctly scaled amplitude spectrum has voltage units. Pressure and sound level require calibration.

The same FFT connects Lab 1 and Lab 2

Lab 1 input: microphone voltage $V(t)$. Peaks identify frequency components in the recorded sound.

Lab 2 input: tracked angle $\theta(t)$. The main oscillation peak gives a period estimate through $T = 1/f$.

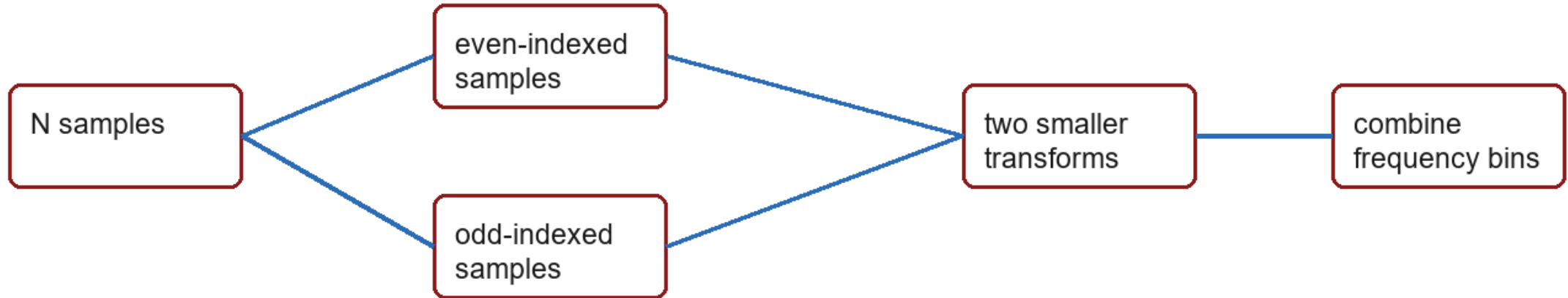
For both: inspect the samples, subtract the mean, use the actual sample rate, and label the spectrum and its units.

Longer records give finer frequency bins: $\Delta f = 1/T_{\text{record}}$. Noise, leakage and aliasing can mislead us. Verify the pendulum peak against the time trace.

FFT algorithm: same DFT, fewer repeated calculations

$$X[k] = \sum x[n] e^{-i 2\pi kn/N}$$

How the FFT makes the DFT fast



FFT gives the same frequency bins as the DFT, but reuses calculations: N^2 work becomes about $N \log_2(N)$.

The DFT compares the record with complex sinusoids. The FFT is an efficient algorithm for computing that same DFT.

It splits the samples into even and odd groups, solves smaller transforms, then combines them.

For Lab 2, MATLAB performs this calculation. You must understand the input, the frequency axis, and the physical meaning of the peak.

Using FFT results in Lab 2

$$f(k) = kf_s/N = k/T_r$$

$$\Delta f = f_s/N_s = 1/T_r$$

Start with the usable theta(t) segment. Remove its mean before the FFT.

N_s samples at frame rate f_s give record duration $T_r = N_s/f_s$ and bin spacing $\Delta f = 1/T_r$.

Read f_{peak} from the largest physically meaningful peak, then use $T_{\text{FFT}} = 1/f_{\text{peak}}$.

A longer usable record narrows the bins. Zero-padding can make a smoother plot but does not improve the true resolution.

Lab 2 trial matrix

Trial A: θ_0 below 10 degrees. Closest test of small-angle theory.

Trial B: θ_0 from 10 to 90 degrees. Observe period change.

Trial C: θ_0 above 90 degrees only when approved by the GTA and the full swing stays in frame.

All releases start from rest. Record θ_0 from protractor; do not push the bob.

Data quality before accepting video

Bob remains in frame.

Bob edge and center are visible without glare.

Pivot and calibration marks are visible.

Release begins from rest.

Quick Review follows bob for nearly all frames.

Notes include file name, angle, frame rate, and calibration distance.

Pre-Lab 2: what you submit

One Word or PDF before laboratory section.

Use the supplied notebook: run the provided cells, keep requested figures, and write your own answers. No programming is assigned.

Task 1: compare point-mass and compound-pendulum frequencies.

Task 2: predict behavior before running the supplied simulation.

Tasks 3-6: explain finite-angle behavior, coordinates, frame rate, and the laboratory plan.

Evidence for combined Lab 2 and 3 report

Schematic: pivot, l , M , m , θ , and positive direction.

Table: measured parameters and analytical, nonlinear, time-domain, and FFT frequencies.

Plots: $\theta(t)$, $\omega(t)$, and spectra with units.

Discussion: separate model limitations from measurement errors.

In-class prediction

For the same apparatus, which release has the longest period?

- A. 8 degrees
- B. 35 degrees
- C. 105 degrees

Explain using the exact sine restoring torque.

Before leaving laboratory

Save and back up every video and MAT output.

Verify tracking for every trial.

Record initial angle, frame rate, calibration distance, l , M , m , and file names.

Check one $\theta(t)$ plot and one spectrum with the GTA.

Key ideas for Lab 2

The real apparatus is a compound pendulum because the bar has mass.

Initial angle changes the nonlinear period.

Camera coordinates require calibration, a fixed pivot, and a verified sign convention.

Compare analytical theory, nonlinear simulation, time history, and spectrum.